Hungi Jung

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Google Scholar ORCID OpenReview LinkedIn

AI researcher with hands-on experience in machine learning, deep learning, and data-driven modeling across medical and clinical domains. Passionate about applying AI techniques to real-world problems, with practical skills in model development, data preprocessing, and explainability.

Education

B.S. in Computer Science, Soongsil University *Mar 2022 – Feb 2025*

Experience

Research Intern, Center for Artificial Intelligence in Healthcare (CAIH) *Sep 2025 – Apr 2026*
Seoul National University Bundang Hospital

- Developed an LLM-based hybrid model for gastric cancer EMR processing
- Designed a structured extraction pipeline for pathology registry variables

Undergraduate Research Intern, PEM Lab, Hongik University *Feb 2024 – Feb 2025*

- Developed an AI model for arrhythmia detection using Python and Colab
- Contributed to signal preprocessing logic and model stabilization
- Presented the research at ACK 2024 (Korea Information Processing Society)

Training

Biohealth AI Academy, AWS × Sungkyunkwan University SAY *Feb 2025 – Sep 2025*

- Participated in AWS official training on cloud computing and AI-based data analysis
- Conducted pre-project and final project using AWS cloud services for biohealth AI

Publications

Structured LLM Pipeline for Automated Gastric Cancer Registry Construction from Bilingual Pathology Reports
Hungi Jung, So Hyun Kang, Myeong ju Kim
Under review, MLHC 2026 Clinical Abstract Track

Wavelet-Based ECG Feature Extraction and CNN for Arrhythmia Classification: An Enhanced Approach Using MIT-BIH Database
YunSeo Jo, Hungi Jung, SeungJu Oh, HaYoon Song
Annual Conference of KIPS, 2024 [Link]

Projects

Extraction and Structuring of EMR Data Using a Query–Regex–LLM Hybrid Model
Sep 2025 – Apr 2026

- Extracted structured clinical data from SNUBH CDW to build registry-ready datasets
- Designed a hybrid pipeline combining regular expressions and LLMs for EMR parsing

- Implemented data transformation logic to standardize heterogeneous clinical records
- Developed a scalable pipeline for automated registry data generation

ML-based Mortality Prediction Dashboard using MIMIC-IV *Jul 2025 – Sep 2025*

- Extracted key features from MIMIC-IV to predict in-hospital mortality across wards, ICUs, and EDs
- Trained XGBoost, Transformer, and TCN models; selected optimal model by performance metrics
- Applied SHAP and LIME for model explainability
- Developed an AI-based dashboard for healthcare professionals

CNN + XAI-based Brain Tumor Detection *Jun 2025 – Aug 2025*

- Utilized an open-access brain tumor dataset from Kaggle
- Designed preprocessing and classification tasks using CNN models
- Applied Grad-CAM for model interpretability

AI-based Arrhythmia Detection Using MIT-BIH Dataset *Jun 2024 – Apr 2025*

- Designed a deep learning model for arrhythmia detection from ECG data
- Built a preprocessing pipeline incorporating cardiac domain knowledge and signal characteristics

Research Interests

Machine Learning & Deep Learning · Explainable AI (XAI) · Medical AI & Clinical Data Analysis
Time-series modeling · Computer Vision · AI systems & deployment

Skills

Languages: Python, Java, SQL
ML/DL: PyTorch, scikit-learn, XGBoost, pandas, numpy, matplotlib, wfdb
Cloud: AWS
Tools: Git, Jupyter Notebook, Google Colab

Certifications

Network Advisor Level 2, ICQA *Jan 2023*